

2920/205
VISUAL PROGRAMMING
July 2017
Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE II
VISUAL PROGRAMMING

3 hours

INSTRUCTIONS TO CANDIDATES

*You should have an answer booklet for this examination:
This paper consists of **EIGHT** questions.
Answer any **FIVE** in the answer booklet provided.
All questions carry equal marks
Candidates should answer the questions in English.*

This paper consists of 5 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

- 1 (a) (i) Define the term *Visual* as used in Visual Programming. (2 marks)
- (ii) Other than Visual Basic. State **four** other Visual Programming languages that could be used for application development. (2 marks)
- (iii) A computer lab technician wants to install a Visual Basic 6.0 on a computer with the following specifications:
- Android Operating System
 - A 32 MB RAM
 - Hard disk space of 116 MB

Explain **two** reasons that may make the installation not successful. (4 marks)

- (b) Outline **four** factors to consider when naming objects in Visual Basic programming environment. (4 marks)
- (c) Figure 1 show an interface used to enter information into a database .Use it to answer the questions that follow:

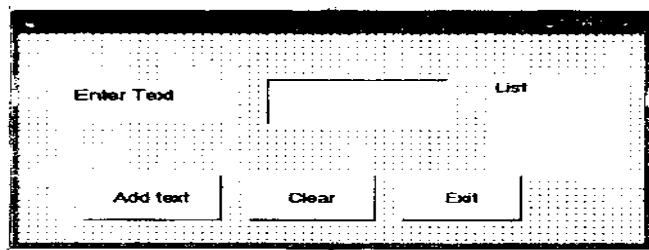


Figure 1

Write a Visual Basic program that would implement each of the command buttons. (8 marks)

- 2 (a) Explain **two** ways used in a Visual Basic program to *pass parameters* to a function. (4 marks)
- (b) Describe the **three** levels of variable declaration in a Visual Basic program. (6 marks)
- (c) Distinguish between *name* and *text* property of an object as used in a Visual Basic environment. (4 marks)
- (d) A soda wholesaler sells a 300ml bottle of soda for Kshs 25 when a customer purchases less than 200 bottles and Kshs 23 per bottle when a customer purchase 200 or more.

Write a Visual Basic program that prompts the user for the total number of bottles purchased through the use of a textbox. The program should then compute the total cost and display the results on a picture box. (6 marks)

- 3 (a) (i) State the type of error to be displayed in each of the following cases: (3 marks)
- (I) Using an If statement without an end If;
 - (II) Program adding two numbers 50+50 and gives results as 5050;
 - (III) Using a prefix *txtfistnumber* instead of *txtfirst* that exists on the property window.
- (ii) Explain a reason that may necessitate the use of each of the following debugging tools in a Visual Basic Program. (6 marks)

(I)



(II)



(III)



- (b) Explain the use of each of the following properties assigned to a label at design time in a Visual Basic program. (6 marks)
- (i) Autosize;
 - (ii) Forecolor;
 - (iii) TabIndex.

- (c) Write a Visual Basic program that would display the following multiplication table on a form when a number 6 is entered in a textbox. Use a *for loop*. (5 marks)

6 X 1 = 6
 6 X 2 = 12
 6 X 3 = 18
 6 X 4 = 24
 6 X 5 = 30
 6 X 6 = 36

- 4 (a) Other than the use of commas, describe **two** functions that may be used to format text in a Visual Basic program. Give an example in each case. (6 marks)
- (b) Distinguish between *data control* and *database engine* as used in a Visual Basic program. (4 marks)
- (c) (i) Outline the use of a boolean flag in programming. (2 marks)
- (ii) A company computes tax from employee's salary as follows:

SALARY	TAX	MESSAGE
Less than 10000	No tax	No tax
10000 and 20000	10%	Lowest tax
21000 and 35000	15%	Middle tax
Above 35000	20%	Highest tax

Table 1

Write a Visual Basic program that would accept salary of an employee through the use of a textbox. The program should then calculate the salary less tax to be paid and the results to be displayed on a label. Use an *If...Then...Else statement*. Attach the code to a command button. (8 marks)

5. (a) Outline **four** navigational methods of a *Recordset* object. (4 marks)
- (b) Describe **three** Optional Parameters of the *OpenDatabase Method* that are used during the connection. (6 marks)
- (c) A programmer intends to display a graphic on a Visual Basic program interface. Describe **two** controls that he would use to achieve his goal. (4 marks)
- (d) Martin would like to use *Recordset* objects to manipulate data in a database. Describe **three** that he is likely to use. (6 marks)

- 6 (a) (i) Outline **three** benefits of using control arrays in a Visual Basic program. (3 marks)
- (ii) Distinguish between a *parameter* and a local *variable* as used in a Visual Basic Program. Give an example in each case. (4 marks)
- (b) Describe **three** circumstances under which ADO cursors may be used during data connection. (6 marks)
- (c) Table 2 shows remarks displayed when a letter is input into a computer. Use it to answer the question that follows

No of letters in a word	Remarks
0	No word entered
1	One letter word entered
2	two letter word entered
3	Three letter word entered
4	More than three letters entered

Table 2

Write a Visual Basic program that would accept a word through the use of an Inputbox function. The program should then determine the number of characters in the word and display the remark on a picture box based on the number of letters as shown in table 2. Use *case* statement. (7 marks)

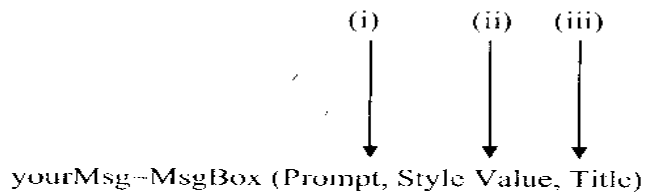
- 7 (a) State **six** examples of databases supported by a Visual Basic program. (3 marks)
- (b) Explain **two** effects of using *option explicit statement* in a Visual Basic program. (4 marks)

- (c) (i) Given the following code segment
- ```
Sub myfirstsub()
 Print Tot (2,3)
End sub

Public Function Tot(x as single, y as single) as single
 Tot =x + y
End Function
```

Interpret the code. (5 marks)

- (ii) The following is a message box function



Outline the function of each of the parts labelled (i),(ii) and (iii) . (3 marks)

- (d) Write a Visual Basic program that would display each of the following lines of text in a multiline textbox when a command button is clicked: (5 marks)
- See no evil*  
*Speak no evil*

- 8 (a) Describe **three** properties that could be used with a *datalist* in a Visual Basic program. (6 marks)
- (b) Distinguish between *local window* and *watch window* as used in a Visual Basic program. (4 marks)
- (c) State a reason why each of the following Visual Basic conditions is wrong. (3 marks)
- (i) temperature >45 and <65
- (ii) value1 and value2=5
- (iii) case is 60 to 69.
- (d) Figure 2 show a form design in Visual Basic intended to perform addition and subtraction. The command button captioned Addition and subtraction all are both named *operator*. Write a Visual Basic program that would compute either addition or subtraction when either button is clicked. The answer should be displayed on a label. Use case statement. (7 marks)

Figure 2

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